

Stage 3 – Additional Questions and Answers

Note

The shared questions are answered in the completed lab PDF. This file answers the remaining questions and adds explanations where the PDF only partly covers them.

Should Database be a class?

Database does not need to be a class in the model. It is a tool for storing information storing than a person or booking represented by the system. Can be handled later when system is later stages of development

Patient to Appointment: which relationship and why?

Patient and appointment have an association, meaning they are linked. One patient can have zero or many appointments. But an appointment needs one patient. I chose this relationship because an appointment needs a patient, but neither is a type of the other class. The requirements do not say that deleting a patient should automatically delete their appointments, so I did not use composition.

Should Appointment inherit from Patient?

No. Inheritance means one class is a type of another class. An appointment is not a type of patient.

Instead, Appointment stores a reference to the Patient attending the booking.

Critique the AI proposal: PatientManager

The existing domain model doesn't need a ClinicController. It is not clearly responsible according to the guidelines.

A controller does not have to be a fundamental domain class, but it could subsequently aid in coordinating operations in the functional application.

Critique the AI proposal: PractitionerManager

A separate PractitionerManager class is not needed. Practitioner can update their own details and hold them. Finding all bookings for practitioners can be handled separately.

Critique the AI proposal: ClinicController

Not needed. Requirements do not give it a clear responsibility, which means the class will be inefficient. It can help coordinate actions, but it doesn't need to be included.

Responsibility allocation

Each class looks after its own information.

Patient and Practitioner manage their own details. Appointment manages its booking time, status and links to the patient and practitioner.

Checking for booking conflicts needs information from other appointments. This should be handled by a separate booking service when the application is developed.

Key relationships

Each appointment links to one patient and one practitioner. Both patients and practitioners can have zero or many appointments.

These links avoid copying patient and practitioner details into every booking. They also make information easier to update.

Details requiring confirmation

The allowed statuses, exact booking conflict rules and cancellation rules still need confirmation. Rescheduling and cancellation operations are proposed and should be confirmed before they are fully implemented.